WELLSENSE · PROBLEM STATEMENT

WellSense problem statement

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-25	Amish Chadha	Kit 1.8 users-and-needs and stakeholder tables added (Tables 2 and 3), the source of the user requirements in WLS-URS-001; requirements now in WLS-PRS-001
0.6	2026-10-02	Amish Chadha	Partner and region answered by Amish's decision of 2026-10-02 (WLS-DEC-001)
0.5	2026-09-26	Amish Chadha	Budget approved by Amish; constraint now $200
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Reflect WLS-DDR-001 (budget covers WellSense parts only) and the TRL 3 cost and access findings of WLS-CAL-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, prior work)
0.1	2026-09-25	Amish Chadha	Initial scaffold

WellSense problem statement

Groundwater is falling in many regions, yet most wells have no measurement, so over-pumping is noticed only when wells run dry. The people who share an aquifer need a cheap, trustworthy record of how the water level in their own wells moves through the day, the season and the years, in a form they can read and discuss.

The problem

Groundwater supplies about half of the water withdrawn for domestic use worldwide, including drinking water for most rural people, and about 25 % of irrigation water (UNESCO, World Water Development Report 2022). A synthesis of in situ records from about 170,000 monitoring wells in 1,693 aquifer systems found rapid declines of more than 0.5 m a year widespread in the twenty-first century, and accelerating declines in 30 % of regional aquifers (Jasechko et al., Nature, 2024). The same study shows recoveries where pumping was regulated or recharge increased, as in the Bangkok basin, so measured decline can be reversed when people act on it.

Most wells that people actually draw from are not monitored. Official networks use a small number of dedicated observation wells, and data reach village councils, farmer groups and handpump committees late or not at all. Owners learn that the aquifer is falling when the pump starts to suck air or the handpump runs dry, and the usual response is to drill deeper. In the United States, new wells are being built deeper 1.4 to 9.2 times more often than shallower (Perrone and Jasechko, Nature Sustainability, 2019), a stopgap that the authors call unsustainable.

Commercial water level loggers exist and work well, but they cost several hundred dollars each before telemetry, need vendor software, and are usually owned by an agency rather than the community. There is no open, low-cost reference design that a community group, a WASH organization or a school can build, check and repair.

Users and context

Table 1. Users and context.

User	Need	Context
Village water committee or farmer group sharing an aquifer	See whether the water table is falling, how fast, and how it responds to pumping and rain; set local rules	Rural India, sub-Saharan Africa, Mexico, Middle East; shared tube wells and boreholes
Handpump or borehole caretaker	Know when static level approaches the pump intake before the supply fails	Community water points, often no grid power
Farmer with a private irrigation well	Plan pumping hours and crop choice; avoid dry wells	Seasonal peaks in the dry season
WASH NGO, utility or groundwater agency	Low-cost network densification between official monitoring wells	Programme monitoring, drought early warning
Hydrogeology student or researcher	Open, inspectable data and hardware for teaching and small studies	Universities, schools

Table 2. Users and needs (kit 1.8; the source of the user requirements in WLS-URS-001).

User type	Goals and needs	Typical use scenario	Pain points and challenges
Village water committee or farmer group sharing an aquifer	A trusted, continuous record of the water table that shows whether it is falling, how fast and how it answers pumping and rain, to set local pumping rules	Monthly meeting looks at the season's curve on a phone and compares it with last year	No measurement until wells fail; official data arrive late; costly loggers belong to agencies
Handpump or borehole caretaker	Early warning when the static level nears the pump intake; simple checks they can do themselves	Quarterly visit: tape reading, desiccant change, glance at the dashboard	No grid power; tampering and livestock at the wellhead; little time and few tools
Farmer with a private irrigation well	Plan pumping hours and crop choice through the dry season; see drawdown and recovery	Runs a pumping test with the logger at 1 min and reads the recovery	Wells go dry at the peak of the season; deepening a well is the usual answer and is costly
WASH NGO, utility or groundwater agency	Low-cost densification between official monitoring wells, with data they can merge	Installs a handful of loggers beside a programme's wells; exports CSV	Commercial loggers cost several hundred dollars before telemetry; vendor software
Hydrogeology student or researcher	Open, inspectable hardware and data for teaching and small studies	Builds one in a lab, compares it with a tape and a commercial logger	Closed instruments cannot be inspected or repaired

Table 3. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
ACWADAM, Pune (first candidate co-design partner, WLS-DEC-001; not yet approached)	Participatory groundwater management with farmer groups on shared aquifers; sets the design well	Co-design sessions, well sizes, site access, field trial at TRL 5
Gram panchayat water committees and farmer groups	Host the wells and own the data	Consent, siting, data-sharing rules, dashboard language
Well owners and pump installers or drillers	Own the pump, riser and wellhead the logger shares	Permission to open the wellhead; advice on fitting the tube beside a pump
Central and state groundwater bodies (CGWB, state groundwater departments, CGWA)	Run monitoring networks and regulate extraction; CGWA's guidelines require piezometers with digital water level recorders for larger abstractors (WLS-REG-001)	Data formats; whether community data can complement official records
WASH NGOs and utilities	Programme monitoring and drought warning	Pilot sites, maintenance arrangements
FieldNode and TwinKit projects (same lab)	Supply the core node and the gateway	Port pinout, energy allowance, 55 degC rating (WLS-DEC-001)
Helpful Engineering volunteers and universities	Build, review and teach with the design	Build reports, design suggestions

Typical wells are 100 to 200 mm (4 to 8 in) boreholes 20 to 150 m deep with a submersible pump or handpump riser inside, or large-diameter dug wells. Static water levels lie from a few metres to more than 50 m below ground, with seasonal swings of several metres in intensively pumped areas. The wellhead may be exposed to sun, flooding, livestock and tampering. Mobile coverage is patchy; the lab's shared node, FieldNode, reports over LoRaWAN to a TwinKit gateway or a public network.

Constraints

- Garage-buildable prototype, $200 USD in WellSense parts per well at a 30 m probe depth, with the FieldNode core costed in the FieldNode repo (WLS-DDR-001, D1, and WLS-DDR-002, both decided by Amish on 2026-09-25; the figure was raised from $190 to $200 by Amish on 2026-09-26). With the surface conduit added under WLS-DDR-002 the parts cost $197.60 at 30 m, so the budget holds to about 31.3 m (WLS-CAL-001 v0.3).
- Built on the lab's shared FieldNode core for power, radio and enclosure; WellSense adds only the probe, wellhead parts and interface.
- No change to the water supply: nothing in the well may contaminate the water, and the well must stay sealed.
- Must work beside an existing pump and riser without pulling them where possible; the probe body is limited to 22 mm so it runs freely in a 1 in access tube (WLS-DDR-002).
- Data belongs to the community that hosts the well. Readings are water levels only; the owner decides what to share.
- Hand tools only for installation; parts available from plumbing, electrical and online suppliers.

Prior work

- **Commercial loggers.** Absolute-pressure loggers with barometric compensation and vented 4 to 20 mA transducers are the standard tools in groundwater monitoring. They are accurate and robust but closed and costly for community use.
- **Standard methods.** The U.S. Geological Survey publishes procedures for measuring well water levels with a steel tape, an electric tape and a submersible pressure transducer (Cunningham and Schalk, USGS Techniques and Methods 1-A1, 2011). WellSense follows the same logic: a transducer for the continuous record and a manual tape for the datum and periodic checks.
- **Open loggers.** The Cave Pearl Project showed that an Arduino-class logger built from breakout boards can run for more than a year on three AA cells for $25 to $50 before sensors (Beddows and Mallon, Sensors, 2018).
- **Community groundwater management.** In the Andhra Pradesh Farmer Managed Groundwater Systems project, farmers took fortnightly measurements of static and pumping water levels in their own wells, ran rain gauges and drew up crop-water budgets before the winter season; about half a million farmers in 661 habitations took part (FAO, *Smallholders and Sustainable Wells*, 2013). WellSense aims to make that record continuous and cheap.

Out of scope

- Water quality (conductivity, nitrate, arsenic). A conductivity probe could share the port later.
- Pump control or metering of abstraction.
- Aquifer modelling beyond simple trend and drawdown charts.

Open questions

- First partner and region for co-design: an Indian farmer group, an African handpump programme or a US groundwater agency? Decided by Amish, 2026-10-02 (WLS-DEC-001): a participatory groundwater management group in India that works with farmer groups on shared aquifers; the first candidate to approach is ACWADAM in Pune, with its well sizes setting the design well and the FieldNode core at its sites on the IN865 band; not yet approached.
- How much access to the wellhead can a community give: can an access tube be added without pulling the pump? WLS-CAL-001 shows a 1 in tube fits beside the riser of a 150 mm well in plan, but only an installation trial can show whether it can be lowered past an installed pump's riser and cable.
- Who holds and publishes the data, and in what language and form do users want the dashboard?

User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [ ] Identify a local partner organization (Helpful Engineering network, NGO or university)
- [ ] Run co-design sessions with intended users; record who, where and what was learned
- [ ] Validate load, distance, terrain and cost assumptions in the field
- [ ] Revise the user and product requirements (WLS-URS-001, WLS-PRS-001) from findings before freezing the design